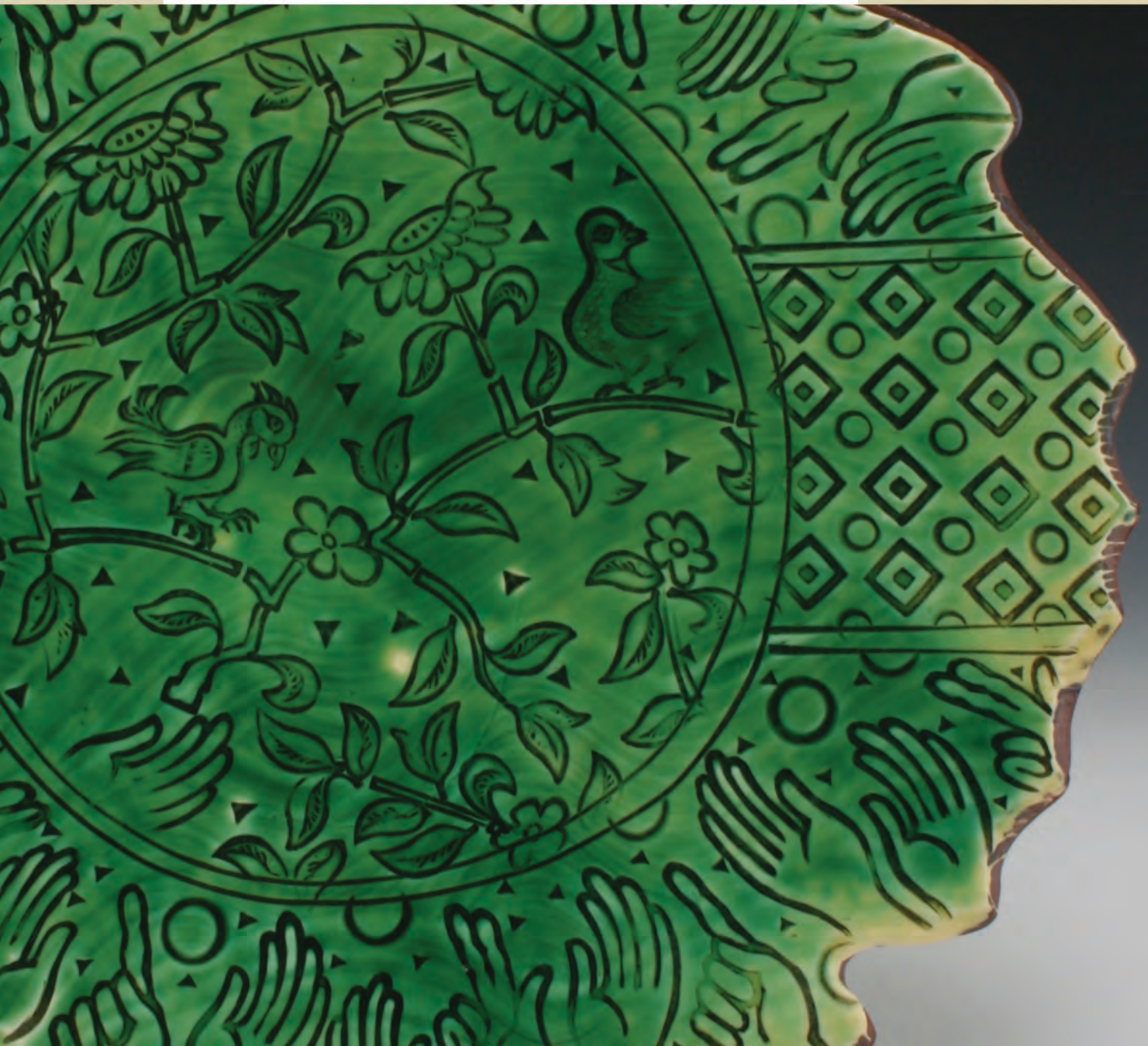


LOW-FIRE GLAZES AND SPECIAL PROJECTS

Edited by Holly Goring

Ceramic
Arts
Handbook
Series



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ISBN: 978-1-57498-378-4 (Paperback)

ISBN: 978-1-57498-588-7 (PDF)

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Publisher: Charles Spahr, Executive Director, The American Ceramic Society

Managing Director: Sherman Hall

Editor: Holly Goring

Graphic Production: Pamela S. Woodworth

Series Design: Melissa Bury

Cover Image: *Plate of Many Hands* by Paul Linhares

Frontispiece: Footed vase, by Joan Bruneau



Table of Contents

1	Bases and Liners	
	Clear Boron Glazes	1
	<i>Lisa Orr</i>	
	Low-Fire Layering	3
	<i>Matt Wedel</i>	
	Low-Fire Liner Glazes	4
	<i>Deanna Ranlett</i>	
	Three Ingredient Glazes	6
	<i>Holly Goring</i>	
	Low-Fire Alkaline Glazes	8
	<i>Cheryl Pannabecker</i>	
	Glossy and Opaque Glazes	11
	<i>Martina Lantin</i>	
	Low-Fire Chun Glaze <i>from Trevor E. Youngberg</i>	13
	Versatile Base Glazes	14
	<i>Matt Ziemke</i>	
	Wood Ash Base <i>from Mark Pharis</i>	16
	Low-Tech Recipes to Test	17
	<i>Lowell Baker</i>	
2	Clays, Slips, and Engobes	
	Preventing Low-Fire Scumming	19
	<i>Lilly Zuckerman</i>	
	White Slip Over Earthenware	20
	<i>Martina Lantin</i>	
	Slip for Sgraffito	21
	<i>Shoko Teruyama</i>	
	Colored Engobes <i>from George Rodriguez</i>	23
	Developing a Patina	24
	<i>Rafael Molina-Rodriguez on the work of Lee Akins</i>	
	Finishing with Thick and Thin Glazes	26
	<i>Magda Gluszek</i>	
	Screen-Printing Slip <i>from Israel Davis</i>	28
	Wide-Range Recipes to Test	29
	<i>Gerald Rowan</i>	



3 Textures and Special Effects

Soda-Fired Earthenware 33

Peter Held on the work of George McCauley

Lichen Glaze 35

Mandy Ginson on the work of Darren Emeneau

Crater and Lava Glazes *from Bob Reed* 37

Low-Fire Crystal glazes 39

Cary Esser

Egyptian Paste 41

Mary Cloonan on the work of Deborah Sigel

Self-Glazing Clays to Test 46

Gerald Rowan

Natural Textures *from Lee Akins* 49

Foaming Fat Glaze 50

Matt Wedel

4 Carving, Incising, and Sgraffito

Carving Bright Slips 51

Kristin Pavelka

Folkware Sgraffito 55

Denise Wilz

Sgraffito Techniques on Terra Sigillata 57

Karen Newgard

Decorating Nesting Bowls 60

Courtney Murphy

5 Working with Slips

Slipware Marbleizing 63

Michelle Erickson and Robert Hunter

Redware Slip Trailing 69

Denise Wilz

Trailed Slip and Translucent Glaze 71

Arthur Halvorsen

Polychrome Surfaces 73

Joan Bruneau



	Slip Image Transfers	76
	<i>Jason Bige Burnett</i>	
	Painting Realism with Slips	81
	<i>Tim Ludwig</i>	
6	Majolica	
	The Wonderful World of Majolica	85
	<i>Linda Arbuckle</i>	
	Majolica Overglaze Techniques	90
	<i>Janice Strawder</i>	
	Watercolor Maiolica	93
	<i>Laurie Curtis</i>	
	Unglaze Majolica Coloring	97
	<i>Courtney Murphy</i>	
	Majolica Decorating with Stencils	101
	<i>Clay Cunningham</i>	
	Overglaze Majolica Coloring	103
	<i>Liz Quackenbush</i>	
7	Combining Techniques	
	Stamping Decoration	105
	<i>Paul Linhares</i>	
	Paper Resist and Pouncing Patterns	110
	<i>Ursula Hargens</i>	
	Accentuating Texture with Stains	114
	<i>Marty Fielding</i>	
	Using Text and Graphic Resist	118
	<i>Connie Norman</i>	
	Constructing a Low-Fire Glaze Surface	121
	<i>Kari Radasch</i>	
	Painterly Surfaces of Lauren Mabry	125
	Crackle Surfaces	127
	<i>Nicole Copel on the work of Yoshiro Ikeda</i>	
	Pairing Pots and Drawings	130
	<i>Bede Clarke</i>	

Preface

First and foremost this collection of low-fire recipes and surface-design projects is meant to inspire those who have a love of making and a passion for materials. Whether you already work in low-fire temperatures or want to add a new range to your repertoire, this book is perfect for you.

When considering whether to work with low-fire clay and glazes, it's important to first dispel any misconceptions, including that low-fire clay is not as strong as high-fired ware and that it isn't suited for functional work. I think Kari Radasch (page 121) addresses this best, "It is a shallow notion to insist that low-fire work lacks the glaze depth that stoneware and porcelains claim to have. On the contrary, it is as much the case that high-fire work lacks the glaze depth of terra cotta. There are many reasons for this misperception, but the most preeminent one, in my opinion, is that our low-fire vocabulary is not as developed as our high-fire one. We will begin to fix that right here." Beyond its well-documented history of Greek and Roman vessels, Southwestern adobe housing surfaces, tin-glazed Delft ware, and Iznik tile, low-fire clay bodies and surfaces have a rich visual weight, a long firing range, and an incredible surface depth. This book is meant to help you focus on both low-fire forms and techniques, as one cannot evolve naturally without the other, and to help you extend your skill range of materials and processes.

What I love about this compilation of articles and recipes, pulled from the best of *Ceramics Monthly* and *Pottery Making Illustrated*, is that it reinforces basic skills of commonly used ceramic practices while also introducing a host of intermediate and advanced techniques from respected artists who are currently practicing their craft just as you are. These tried-and-true techniques will teach you everything from developing a patina on your sculptural surfaces to marbling slips on large functional platters, from creating traditional pouncing patterns with graphite to making image transfers with underglaze on newsprint.

This book is also meant to encourage you to get into the studio and test new recipes, from simple three-ingredient glazes to lichen and lava glazes, which give dramatic results. Not only are there an abundance of recipes to test, but there is also information for mixing and applying slips, terra sigillata, engobes, majolica glazes and overglazes, Egyptian pastes, as well as glazes ranging from liners to mattes and alkalines. Keep in mind your results may vary from what is pictured here due to variables such as material origins, clay bodies, mixing styles, firing temperatures, cooling rates, water supply, and of course, application methods. Testing glazes is a chance to create something new rather than to simply repeat. Combine a project from one section with a glaze from another. Make each recipe and each technique your own. That, of course, is the best part of ceramics after all.

Holly Goring

Safety

When mixing glazes, safety should be your main concern. Every supplier is required to keep Material Safety Data Sheets (MSDS) on hand for every material used in a glaze. These contain safe handling procedures and any toxicity warnings. Wearing a NIOSH approved and professionally fitted respirator, safety glasses, and dedicating clothing for the studio will lessen your risks to hazardous exposure. Store materials in plastic containers or approved bins out of reach of children and pets. Heed all warnings for the proper handling of the materials.

Bases and Liners

LOW-FIRE LINER GLAZES

by Deanna Ranlett

What is a liner glaze? Why do you need one? A liner glaze is used on the inside of functional pottery to prevent exposure of the user to glaze ingredients that are prone to leaching. In theory it doesn't contain any toxic ingredients such as barium, lead, or heavy metals. It also shouldn't craze. It's a potter's responsibility to put a safely glazed pot into the world that can be used repeatedly with any food without causing harm.

Food-Safe Tests

It's important to test liner glazes to ensure that the glaze inside is a good functional choice and won't fade or etch over time due to exposure to acids commonly found in food or drink (i.e. coffee, lemon, orange, tomato) or bases (i.e., dishwasher soap).

A liner glaze should not contain heavy metals such as manganese, copper, or cobalt, which are prone to leaching. This isn't to say that you can't have colored glazes that pass leach tests—some liner glazes can be colored with small amounts of commercial stains or rutile to create a soft color. If the colored glazes pass the leaching test, you can use them as liners.

Every potter should test for acid leaching in their glazes either with a lemon slice (very acidic and acts quickly) left on the glaze-fired surface overnight or by soaking it in vinegar for 1–2 days. If you see any change in gloss level or color, your glaze is not acid safe and should not be used as a liner glaze. You should also test for deterioration to the glaze through repeated base exposure. Dishwasher detergent is a common base and, over time, it can cause the glaze to wear down. This test

takes additional time, normally requiring the pot to remain in the dishwasher for 30 or more cycles to create any possible change in the glaze.

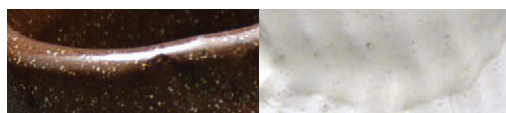
Liner Considerations

Beyond leaching concerns, liner glazes can be beneficial in many ways. For example, using a white or clear liner glaze saves money on dry glaze materials because you won't be including the more expensive oxides and colorants.



Consider glazing the inside and outside of the pot in different colors and on different days. We recently had a workshop where a potter described how much improvement he had seen in his glazing because the clay had dried thoroughly between applications and had therefore accepted the glaze in a smoother, thicker application than when he glazed it all at once.

A white liner glaze highlights what is in the mug—sometimes when I drink tea in a dark mug it appears a bit murky and a nice white or soft colored glaze sets it off better. We tested only clear or white glazes for that reason.



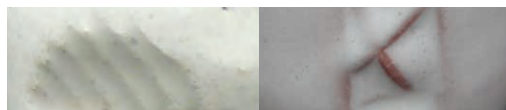
DIGITALFIRE CLEAR (1)

Cone 06–04

Ferro Frit 3134	40 %
Ferro Frit 3124	40
EPK Kaolin	15
Silica	5
	100 %

Add: Bentonite 1 %

The HWC Red Earthenware was fired to cone 02. The 105G Standard Ceramics white clay was fired to cone 04. Brushed and dipped well and was very clear with no bubbles or crazing. Add 6% Zircopax for a nice white.



PETE PINNELL BASE (3)

Cone 04

Ferro Frit 3124	70 %
Strontium Carbonate	10
Nepheline Syenite	10
OM4 Ball Clay	10
	100 %

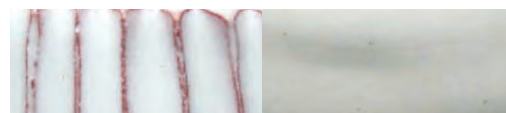
Add: Zircopax 5 %
Bentonite 2 %

The addition of 5% Zircopax creates a nice warm white. This glaze brushed and dipped nicely.

Glaze Recipe Testing

We tested several low-fire recipes with good results. We tested the cone 04 recipes on 105G Standard Ceramics white clay and on Highwater Clay's Red Earthenware. We chose these clay bodies because of the light speckling they each contain, which gives a good indication of the liner glaze's overall coverage. The red clay also provided a nice contrast for the white glazes.

All tiles were tested for acid leaching and none resulted in a change to the glaze's surface. Because firing conditions and clay bodies vary widely—we recommend doing your own tests.

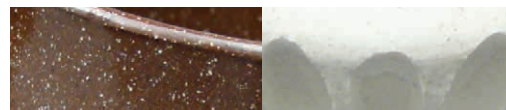


SIEBERT MAJOLICA (2)

Cone 04

Ferro Frit 3124	76.6 %
EPK Kaolin	12.7
Silica	10.7
	100.0 %
Add: Zircopax	10.0 %

This is a wonderful majolica without tin. It makes a fabulous liner—even covering some of the speckling in the 105G Standard Ceramics white clay.



RON ROY CLEAR (4)

Cone 04–02

Ferro Frit 3195	30.0 %
Ferro Frit 3134	30.0
Whiting	6.0
OM 4 Ball Clay	17.5
Silica	16.5
	100.0 %

Add: Bentonite 2.0 %

This glaze is nice to dip and brush. Fired to cone 04 on the 105G Standard Ceramics white clay with great results. On the HWC Red Earthenware the glaze was slightly hazy at cone 04 and better results at cone 02.

Carving, Incising, and Sgraffito

SGRAFFITO TECHNIQUES ON TERRA SIGILLATA

by Karen Newgard



Karen Newgard uses a sgraffito technique in combination with porcelain and terra sigillata to discover a crisp graphic quality with her drawings.

Terra sigillata is like a tight second skin on the pot, often used to seal porous low-fire clay bodies. It's made by deflocculating a clay (or combination of clays) and isolating the fine particles. The recipe I use doesn't require ball milling due to the fine clay particles that make up Grolleg kaolin. I find I get a large yield and this recipe is very easy to make. Because water isn't the same in every city, it's critical to use distilled water to maintain consistent results. Using a drill and paint mixer attachment in a 5-gallon bucket is the easiest method for mixing. Let the mixture sit untouched for 36

hours in a glass container if you have one. After settling, three layers will form: the waste layer on the bottom, a thin layer of water on top, and the usable terra sigillata in the middle. The deflocculant, trisodium phosphate (also known as TSP and available in home stores), charges the particles in the Grolleg causing them to repel and separate. Because the waste at the bottom is so thick, the layers are easy to distinguish, making the decanting process easier. A batch of this recipe lasts me about three weeks. Once mixed and decanted, the



After applying terra sigillata to the surface, lightly draw your design on top of the sigillata using a pencil.



Use a worn needle tool to outline the images. Don't use a sharp tool, it will create a jagged line.



Remove slip from the negative spaces using larger loop trimming and carving tools.



Carving the negative spaces with specific patterns and curving marks adds a sense of rhythm and movement.

terra sigillata should be the consistency of whole milk. I add 1 cup of terra sigillata to 1 teaspoon of black Mason Stain 6600. Always test other stains or oxides before using. If evaporation occurs during use, add a little distilled water. Keep the mixture covered when not in use.

Once a pot is bone dry, brush on two thin coats of terra sigillata, being careful not to overlap brush strokes, and waiting a few minutes between coats. If too many coats are applied it could cause the sigillata to craze and pop off after the glaze firing.

Sgraffito

Use a dull pencil to lightly lay out your design, being careful not to press down too hard because it will leave a mark or remove some of the terra sigillata (1). The goal here is to draw on top of the sigillata, not start the carving process.

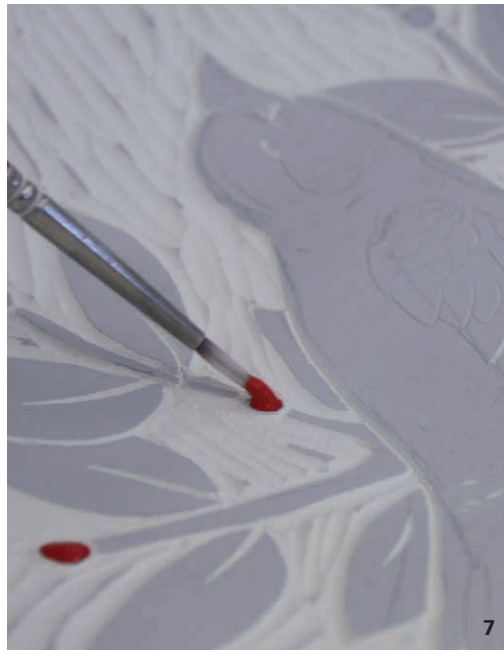
After you're satisfied with the drawing, use a worn needle tool (or similar tool) to outline images (2). If the tool is too sharp, it leaves a jagged line. To remove the background, I use the Kemper WLS carving tool or wire loop tool for tight spaces, and



Carving concentric circles reminiscent of lines created in rippling water and that define banded areas around the rim.



Carving linear patterns around the rim of a bowl that reference a field of crops planted in rows.



Add colored accents to the finished, carved main design or image using commercial underglazes.

for larger spaces I use a small Dolan loop tool (3). With my carving marks I can create a sense of rhythm; making a leaf look like it's blowing in the wind (4) or a bird fluttering in the trees.

In setting up my decorations, I like to give myself starting and stopping points, such as a distinct foot or visible spout and handle attachments.

TERRA SIGILLATA

Grolleg Kaolin	4000 g
Trisodium Phosphate (TSP)	40 g
Distilled Water	28 cups

Mix for 5 minutes and let settle for 36 hours. Decant the middle layer for use. For color, add 1 teaspoon of commercial stain per 1 cup of terra sigillata.

Conversely, I often negate the form all together and wrap a scene around the piece.

Repeated patterns have always been appealing to me; repetition of a field of wallpaper motifs, lines created in rippling water, (5) or crops in rows (6). Looking for these types of arrangements in nature or our modern landscape is a source inspiration.

Caution: The carving process produces a lot of dust so I wear an apron and carve over a square of egg-crate foam that catches the shavings, keeping them contained and making them easy to dispose of. The dust created in this process is extremely fine. Never blow the dust away while working, and never sweep it away. Only use a only vacuum with a HEPA filter.

When the carving is complete, I add a pop of color with AMACO Velvet underglazes to accent the main images (7).

7

Combining Techniques

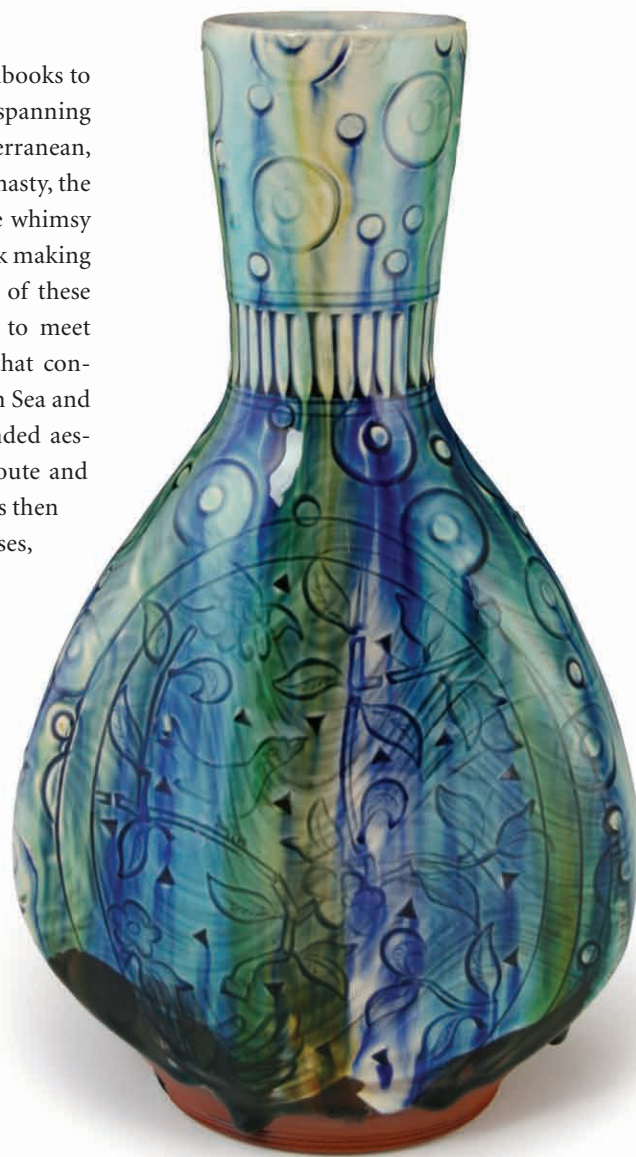
STAMPING DECORATION

by Paul Linhares

Looking back through years of work and sketchbooks to find the common threads, I saw inspirations spanning the energetically patterned work of the Mediterranean, the fluid glaze and intense color of the Tang Dynasty, the grace and pure forms of the Song Dynasty, the whimsy of Medieval slipware, and the spontaneous mark making of Korean hakeme. As I investigated the roots of these disparate influences I found they all seemed to meet around the 9th century along the Silk Road that connected China and the East to the Mediterranean Sea and the West. This ancient melting pot had a blended aesthetic that affected all of the cultures on its route and many that came after. Having found a root, I was then free to explore these influences with new processes, patterns, and meanings every day in the studio.

Applying Slip

Who doesn't love thick, sloppy slip covering a pot? Slip consistency will determine the opacity and texture on the final piece. For brushing, thin the slip just to the point that it starts to drip off the brush. Use a soft, wide bamboo hake brush for scooping out the slip and smearing it on the pot (1). Cover as much of the surface as possible before it starts to dry, rolling the brush to get the slip off quickly. Then spin the pot on the wheel while adding extra slip to even out the coating. Next, use a cheap, plastic-bristle brush to leave a pronounced brush stroke texture that occasionally scrapes down to the clay underneath (2). Leave some of the pot unslipped to reveal part of the process and let the clay body show through for contrast.



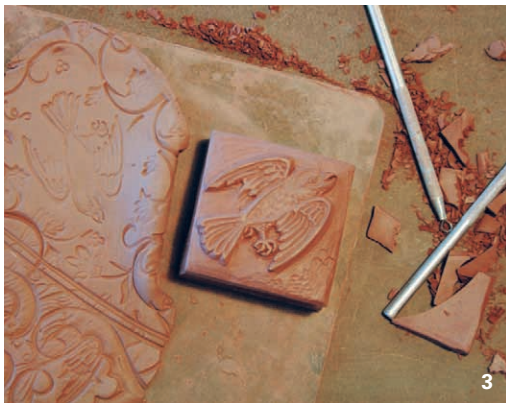
Paul Linhares discovered a common thread in his years of research into various traditions that led him to develop a blended aesthetic.



Cover the surface with a thick coat of white slip using a bamboo hake brush. Leave the base bare



Finish the slip coat using a coarse plastic-bristle brush to create a brush stroke texture.



Carve multiple stamps from leather-hard clay, allow them to dry, and bisque fire them.



Use a handmade roller to create defining borders and decorative lines.

Drawing with Stamps

Around 2004 I began working with wood blocks to press shapes into my pots. Shortly after, I came across a book with several pages of Persian clay stamps while researching Islamic patterns. I realized that I could draw designs with impressed pattern if I only had enough stamps.

I began carving leather-hard clay (think Parmesan cheese consistency) with an X-Acto knife and mini loop tools into a library of stamps (3). To make a stamp set for my thistle pattern, (based on the Iznik poppy), I carved a flower head, a straight stem, a slightly curved stem, and a hard curved stem into different faces of a clay block, plus large and small leaves facing left and right. Once dry, these were bisque fired. I also use the sanded edge of a wooden

rib for continuous straight lines and wheel-thrown clay rollers for continuous curves.

When planning my surface designs, I play back and forth between the wide-open, all-over-the-pot style of the Bronze age Minoans and the carve-up-the-form-into-picture-planes tendencies of the Renaissance Italians, using a contemporary lens that is bent on disrupting that classical visual hierarchy. At least that's the kind of thinking going on in the background when my intuition takes over and does the hard work of actually deciding where things go.

When the slip layer is dry enough not to stick to the stamps, start rolling on borders with a tool made from a notched piece of wood and a wheel-thrown clay disc held in place by a brad nail (4). This homemade clay roller matches the impres-



Roll stamps on from edge to edge or in a circular motion for a good impression.



Add the branch, leaf, and flower stamps to create a dynamic composition.



Unify the design by imprinting triangles or similar shapes with the corner of a wood block or the tip of a rib.



Finish the stamping with a geometric pattern to fill in blank spaces and balance out the visual imagery.

sion and edge quality of the clay stamps. Be happy with where the rolled lines end up because there is no eraser. Next, start pressing in your design with your stamp set (5). On a closed form the stamping can't be supported by a hand on the inside, so take care to roll the stamps on from edge to edge or in a circular motion to get a good image without denting the pot too badly. Sometimes it doesn't work and much of the detail of a larger stamp will be lost in the middle. Selecting smaller stamps for closed forms can resolve that problem. I add branches, leaves, and flowers with the goal of a balanced and dynamic composition (6), then fill the little empty spaces with triangles made by pressing the corner of a wood block into the clay (7). The triangles make a deep

mark and turn out as dark points that act as a unifying element in the piece. Finally I press in a circular-shaped stamp in the areas surrounding the design—the geometric pattern balances out the organic images (8).

After the pot is decorated, roll the bottom edge, then smooth it with your thumb. Allow the piece to dry, then bisque fire it.

Brushable Colors and Glazing

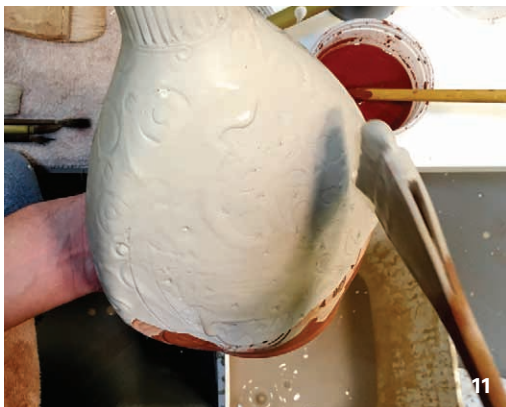
I want the user of my pottery to feel the dents caused by the stamping process on the inside of a mug, see the motion of the slip in the marks made by the brush, and have no doubt when they pick up one of my pieces that the glaze is a coating of melted glass. I have always been drawn to the ex-



Sponge the Brushable Color Mix into the impressions left by the stamps and the bristle brush.



Remove as much color as possible without wiping too much out of the texture.



Even out the poured glaze coat with a brushed coat of the same glaze.



Brush on lightest colors first, being careful not to touch any wet areas.

cessive flux of the Tang Dynasty tri-color glaze. When I was first trying to achieve these colors on my own work it took me a while to realize I was attempting to create lead-glaze effects without using lead. Omitting the lead can cause blisters in what is essentially an over-fired glaze. Along with the brightest copper blue you also get a soft glaze due to the low alumina and high soda required. Additionally, all that soda is soluble even in the fritted form, causing a short lifespan for the glaze in the bucket before it starts to fire strangely. These glaze issues have actually led me to the process I now use to apply color to my pots, in the end adding to the quality of my current work.

Start by brushing or sponging one of the Brushable Color Mixes into the impressions left by the stamps (9), then wipe off the excess with a sponge (10).

Tip: Squeeze it into a tub of water set aside just for that color so it can be reused in the next batch of color mix. Finish with a clean-water wipe down to remove as much color from the pot as possible without wiping too much out of the texture.

Let the piece dry before dipping or pouring on a clear or transparent glaze on the inside and outside of the pot. Before the first coat dries, brush on another coat to even out the glaze (11).

After these base coats dry, brush on the Brushable Color Mixes as if you were decorating majolica ware, but keep in mind how much the glaze will flow (12). NC-14 Clear glaze begins to melt before cone 08, matures by cone 04, and can be fired to cone 1 for a tighter low-fire body. This glaze likes to run so make sure you protect your kiln shelves with kiln wash or fire each piece on its own tile.



Paul Linhares' blue and white yunomis, to 4 in. (10 cm) in height, earthenware, white slip, stamped pattern, glaze, fired to cone 2 in oxidation, 2014.

WHITE SLIP

Cone 04–10 Oxidation

Custer Feldspar	25 %
C and C Ball Clay	25
EPK Kaolin	25
Silica (200 Mesh).	25
	100 %
Add: Zircopax	10 %
Rutile	1 %

Add ¼ cup CMC gum solution per 1000g of water. Thick slips that are meant to give the work some texture lose a little punch in the drying process without some added CMC gum. Mix powdered CMC gum to 1 gallon (3.8 liters) really hot water and let sit overnight.

NC-14 CLEAR

Cone 04–1 Oxidation

Gillespie Borate	15 %
Lithium Carbonate	7
Ferro Frit 3110	40
EPK Kaolin	10
Silica (325 Mesh).	28
	100 %
Add: Bentonite (325 Mesh)	2 %

Antique White:

Add: Red Iron Oxide	4 %
-------------------------------	-----

Add ¼ cup CMC gum solution per 1000g of water. I use CMC gum in slips and glazes to increase fluidity of the slurry and to add dry strength to raw glazed ware. Mix powdered CMC gum to 1 gallon (3.8 liters) really hot water and let sit overnight.

BLUE MIX

Cone 04–10 Oxidation

Cobalt Oxide.	5 g
Red Iron Oxide	4 g
Manganese Dioxide	1 g

I started making this colorant blend after a visit from the Chinese blue-and-white porcelain painters from the Jingdezhen Ceramic Institute. They told us how to make the authentic color and this is my version. I pre-mix this as a dry compound and add it like a regular colorant to my recipes.

BRUSHABLE COLOR MIXES

Cone 04–1 Oxidation

Blue:

Add: Blue Mix	10 %
NC-14 Clear	90 %

Green:

Add: Copper Carbonate	20 %
NC-14 Clear	80 %

Brown:

Add: Red Iron Oxide	50 %
NC-14 Clear	50 %

Mix up a 1000-gram dry batch of the NC-14 Clear base to have on hand for blending in the proportions above.



Holly Goring is Editor of *Pottery Making Illustrated*, where she guides the editorial content and focus of the magazine that states its purpose as being "your resource for ceramic techniques." Goring holds a BFA degree in ceramics from the University of Minnesota-Twin Cities, and maintains a ceramics studio at her home in Columbus, Ohio.

We have high expectations for low-fired work, and that comes from so many great artists pushing the limits of what is possible at the lower temperature ranges. They have tested extensively to find the right fluxes to get just the right effect; they have developed techniques for applying layers of surface to achieve depth and complexity. The recipes and projects in this book come from professional artists who share their knowledge and expertise in order to help make your work the best it can be.

While one of the advantages of working in the lower range of firing is that it uses less energy, perhaps an even more satisfying perk is faster firings and quicker turn-around for testing and experimentation. Smaller kilns are available for this range, making smaller production cycles for the home studio possible. So mix up some glazes, try some of these techniques, and discover new possibilities in no time!



ISBN: 978-1-57498-378-4



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